

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: III

Subject Code & Name: BTCVC304

Hydraulics-I

Max Marks: 60

Date: 06/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define following fluid properties with short explanation:- specific weight, specific volume, specific gravity, viscosity, surface tension, capillarity.	CO2	6
B) State and prove Pascal's law with neat diagram.	CO2	6
C) Define buoyancy and metacentric height. Derive an expression for the metacentric height of a floating body.	CO2	6
Q.2 Solve Any Two of the following.		12
A) The velocity components in a two-dimensional flow field for an incompressible fluid are expressed as $u = 2x + \frac{y^3}{3} - yx^2 ; v = -2y - \frac{x^3}{3} + xy^2$	CO1	6
(a) Obtain an expression for stream function Ψ		
(b) Obtain an expression for velocity potential Φ		
B) Derive Euler's equation of motion. Explain how this is integrated to get Bernoulli's equation.	CO1	6
C) Discuss briefly the different methods of drawing the flow net.	CO1	6
Q. 3 Solve Any Two of the following.		12
A) Derive an expression for mean velocity for laminar flow (i) through a pipe, (ii) between two parallel plates-both plates at rest.	CO4	6
B) Write a short note on Nikuradse's experiment.	CO4	6
C) Explain the boundary layer theory.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain methods of dimensional analysis.	CO4	6

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| B) Explain different types of similarities that must exist between a prototype and its model. | CO4 | 6 |
| C) Explain the terms: distorted models and undistorted models. What is the use of distorted models? | CO4 | 6 |

Q. 5 Solve Any Two of the following. **12**

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| A) What is syphon? Explain on what principle it works? | CO3 | 6 |
| B) Discuss the concept of Water Hammer and Surge Tank. | CO3 | 6 |
| C) Explain the Nomogram and Moody's chart. | CO3 | 6 |

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